

Anatomical scale and architecture organize load-dependent sacroiliac motion

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Abstract

Pelvic mechanical response reflects anatomical variation expressed through a specified load path. We examined sacroiliac joint responses in 278 imaging-derived pelves under five standardized loading configurations and in matched geometry–bone-density variants. Redistributing the same 800 N total acetabular force from bilateral to unilateral support amplified motion and asymmetry; transverse force pairs applied at different sites produced distinct component patterns. Explicit pelvic dimensions substantially attenuated age-adjusted sex-associated translation differences under localized loading, with all fully adjusted confidence intervals spanning zero. Preserving individual geometry while standardizing the bone-density field retained subject-specific translation closely (RMSE ≤ 0.028 mm; minimum identity-line $R^2 = 0.946$). In retrospective adjusted log–log analyses, all 20 sex-specific volume–motion slopes were negative after multiplicity correction. Bilateral standing-like slopes were numerically near simple fixed-force dimensional reference values, whereas several localized-load estimates were numerically steeper. These conditional relations describe size-dependent response under identical prescribed forces across subjects within each load case. Together, the comparisons support a description of pelvic mechanics organized by anatomical scale and individual architecture, with load path determining their mechanical expression. Sex-associated contrasts varied with anatomical adjustment and loading configuration. Interpretation remains conditional on passive static mechanics, the implemented density–modulus law and fixed cartilage, ligament and pretension parameters.

Keywords: sacroiliac joint; pelvic functional anatomy; pelvic biomechanics; sexual dimorphism; pelvic architecture; load transfer; finite-element analysis

1 Introduction

The sacroiliac joints (SIJs) transmit load between the spine and lower limbs through a pelvic ring stabilized by articular geometry, ligaments and muscular compression (Snijders et al., 1993; Vleeming et al., 2012; Hammer et al., 2013). Cadaveric experiments with bodyweight-adapted loading measured small, three-dimensional SIJ movements and substantial interindividual variation (Hammer et al., 2019). Joint motion therefore provides a functional endpoint for examining how pelvic anatomy responds to a specified load.

Pelvic variation spans overall size, inlet and midpelvic proportions, pubic arch geometry and local SIJ morphology. Sex differences in obstetrically relevant dimensions develop along distinct age-related trajectories (Huseynov et al., 2016). Pelvic shape also covaries with stature (Fischer and Mitteroecker, 2015). Fischer and Mitteroecker (2017) distinguished a largely stature-related component of sexual dimorphism in pelvic height-to-width proportions and iliac orientation from predominantly non-allometric differences in the subpubic angle. At the joint itself, analysis of Japanese skeletal specimens identified sex differences in iliac auricular surface size and proportions (Nishi et al., 2020). These findings motivate separate consideration of size, specific dimensions and sex in mechanical comparisons.

The mechanical consequences of pelvic form are relevant to both load transmission and obstetric anatomy (Pavličev et al., 2020). Finite-element models of the pelvic floor showed increasing deflection and tissue stress with increasing floor size under prescribed pressure (Stansfield et al., 2021). For the bony pelvic ring, the corresponding question concerns the relationship between its dimensions and joint motion under prescribed forces. Anatomical covariation with stature and mechanical response to standardized loading address distinct aspects of size dependence.

Evidence for sex-associated SIJ motion comes from computational, imaging and cadaveric studies. Joukar et al. (2018) found greater mobility in a female than in a male lumbopelvic finite-element model under identical loading. Upright MRI in 19 healthy young adults showed greater sagittal SIJ rotation in women during the change from supine to standing (Tani et al., 2023). A study of 24 cadavers likewise found greater range of motion in females, while measured auricular surface dimensions did not distinguish groups with high and low mobility (Shin et al., 2024). Population models permit assessment of how sex-associated contrasts change after adjustment for pelvic volume and dimensions, and how these anatomical associations vary with loading

configuration.

Computational studies also connect anatomy and material representation to SIJ mechanics. Population modeling linked auricular surface area and cartilage volume to local stress, strain and strain energy (Henyš and Hammer, 2025). Other models identified sensitivity of SIJ response to ligament pretension (Heyland et al., 2025) and increased motion following reductions in bone stiffness used to represent osteoporosis (Fan et al., 2026). Matched geometry–density comparisons address the contribution of the represented bone-density field to individual motion predictions. Retaining subject-specific geometry with a standardized density field, and individual density on a common geometry, permits direct comparison with the full subject-specific model while holding soft-tissue properties fixed.

Here, we examined SIJ response in 278 imaging-derived pelvises from the BoneDat workflow (Henyš and Kuchař, 2025) under five standardized loading configurations. Bilateral and unilateral acetabular loads redistribute a fixed total force; localized force pairs probe different application sites on the ring. These are controlled mechanical probes. Childbirth models additionally represent fetal passage and maternal soft-tissue deformation (Chen and Grimm, 2021). We evaluated conditional volume–motion associations with other pelvic dimensions held fixed, changes in sex-associated coefficients after anatomical adjustment, and preservation of individual responses under standardized bone density. All subjects received identical prescribed forces within each load case. The analyses were retrospective; the conditional scaling analysis was exploratory.

2 Materials and methods

2.1 Cohort and provenance

The precomputed cohort comprised 278 pelvic models (150 female, 128 male), aged 16–91 years (mean $\pm$ SD: 53.9 ± 19.3 years), using the BoneDat morphology workflow (Henyš and Kuchař, 2025). Demographic, morphometric and size tables were joined by subject identifier. Eight morphometric variables and the kinematic outcomes were complete for the included cohort (Table 1). Ethical approval was granted by University Hospital Hradec Králové, reference 202411IO3P, with waiver of informed consent.

Table 1: Demographic and morphometric characteristics. Values are mean $\pm$ SD; dimensions are measured from mapped landmarks.

Variable	Overall	Female	Male	Missing
N	278	150	128	0
Age (years)	53.90 $\pm$ 19.33	53.36 $\pm$ 19.74	54.52 $\pm$ 18.91	0
Inlet AP diameter (mm)	117.04 $\pm$ 10.52	121.04 $\pm$ 10.02	112.34 $\pm$ 9.07	0
Biacetabular width (mm)	125.81 $\pm$ 8.53	129.08 $\pm$ 8.21	121.98 $\pm$ 7.22	0
Biischiadic width (mm)	104.27 $\pm$ 9.33	109.87 $\pm$ 7.13	97.71 $\pm$ 7.06	0
Bituberous width (mm)	132.24 $\pm$ 9.15	136.46 $\pm$ 8.29	127.30 $\pm$ 7.49	0
Iliopectineal eminence width (mm)	102.38 $\pm$ 7.75	106.20 $\pm$ 7.02	97.90 $\pm$ 5.96	0
Pelvic inlet transverse (mm)	131.58 $\pm$ 8.88	134.85 $\pm$ 8.59	127.75 $\pm$ 7.61	0
Sacral width (mm)	117.08 $\pm$ 6.08	116.75 $\pm$ 5.67	117.47 $\pm$ 6.54	0
Subpubic angle (deg)	78.12 $\pm$ 10.12	85.80 $\pm$ 6.14	69.11 $\pm$ 5.26	0
Global scale factor (-)	0.96 $\pm$ 0.06	0.92 $\pm$ 0.05	1.00 $\pm$ 0.05	0
Total pelvic volume (cm ³)	823.58 $\pm$ 149.32	736.79 $\pm$ 105.99	925.28 $\pm$ 126.99	0
Total pelvic surface (cm ²)	1568.63 $\pm$ 169.26	1479.09 $\pm$ 133.64	1673.57 $\pm$ 144.80	0

2.2 Anatomical dimensions and their landmarks

Figure 1 shows all eight morphometric measures in Table 1, using the archived template landmarks from the measurement workflow. Each of the seven linear dimensions is the three-dimensional Euclidean distance $d = \|\mathbf{p}_2 - \mathbf{p}_1\|$, rather than its projection into a display plane. The subpubic angle uses the second of three landmarks as its vertex:

$$\alpha_{sub} = \arccos \left[\frac{(\mathbf{p}_1 - \mathbf{p}_2) \cdot (\mathbf{p}_3 - \mathbf{p}_2)}{\|\mathbf{p}_1 - \mathbf{p}_2\| \|\mathbf{p}_3 - \mathbf{p}_2\|} \right], \quad (1)$$

with numerical clipping to $[-1, 1]$ and conversion to degrees. The landmark labels are local to each measurement. Landmarks were mapped from the template to each subject; interpolation settings and archived numerical checks are given in Appendix A.1. Supplementary Figure S2 illustrates the three rotational coordinates, the definition of point translation, and the three bone bodies included in volume and surface-area summaries.

2.3 Finite-element model

The archived simulation metadata specify DOLFINx 0.10, 166,808 cells and 48,291 mesh nodes. Small-strain linear elasticity was solved with PETSc on a shared reference domain. Subject anatomy was established by bijective anatomical registration and represented on the FE discretization by $\mathbf{x} = \mathbf{X} + \phi_i(\mathbf{X})$, using radial basis function (RBF) interpolation of the established registration displacement map. Spatial gradients and integration measures were pulled back to the reference domain (Bonet and Wood, 2008). This interpolation approximates individual

anatomy on a shared discretization and does not reproduce cortical boundaries exactly.

Experimental relationships between bone density and stiffness motivate density-dependent material assignment (Keller, 1994). The archived implementation used the following bone-modulus approximation:

$$E/\text{MPa} = \begin{cases} 2398, & \rho_{\text{ash}} \leq 0.486, \\ 10200\rho_{\text{ash}}^2, & \rho_{\text{ash}} > 0.486, \end{cases} \quad \rho_{\text{ash}} = (\rho + 0.09)/1.14, \quad (2)$$

where numerical density values are in g cm^{-3} and ρ denotes the mapped hydroxyapatite-density input. The implemented approximation comprises this equation, density conversion and low-density floor. The constant low-density branch suppresses modulus variation below its threshold. Bone Poisson ratio was 0.30. Both SIJ cartilage and pubic symphysis used $E = 1.0$ MPa and $\nu = 0.45$, as recorded in all three simulation archives. These nominal choices were not calibrated to individual subjects.

Ligaments were assembled as linear spring stiffness matrices with a geometric stiffness contribution from prescribed pretension and a pretension force vector. The implementation does not update a tension-only active set during the static solve. Bilateral anterior SIJ, posterior SIJ, interosseous, sacrospinous and sacrotuberous bundles and the pubic bundle were included. Per-bundle stiffnesses were respectively 1400, 5600, 200, 3000, 3000 and 1000 N/mm; total pretension was 20 N per bundle. Each bundle’s parameters were divided among its segments. Bone–cartilage interfaces were conforming; frictional contact and muscle activation were not modeled.

2.4 Loads and constraints

The superior S1 facet was constrained using a surface penalty $\gamma = 10^6$ N/mm³ (Appendix A). Figure 2 shows the load application sites; Table 2 gives force vectors in the simulation’s global coordinates; these must be distinguished from the subsequently constructed pelvis-aligned reporting frame. SP2leg applies +400 N vertically to each acetabulum, and SP1leg applies +800 N to the right acetabulum. LAB1 applies opposing ML forces of ± 400 N to the internal ring surfaces; LAB2 applies the same signed pair to the ischial tuberosities; LAB3 applies opposing AP forces of ± 400 N at the pubis and sacrococcygeal junction. Inspection of the archived template places both left application patches at greater global x than their right counterparts; thus both

ML pairs point outward in that template, consistent with outward loading at both sites. We name the cases by application site because a force direction does not establish the direction of joint motion.

Surface tractions were normalized by mapped surface area, giving identical prescribed force resultants across subjects. LAB force pairs have zero net force, whereas standing cases have an 800 N net vertical force balanced by the sacral constraint. Equal sums of force magnitudes do not imply equal moments, work, or physiological demand. The analyses compare responses to these standardized mechanical probes. All load cases include the prescribed ligament pretension.

Table 2: Standardized loads in global simulation coordinates. LAB1–LAB3 are independent mechanical proxies.

Load case	Mechanical proxy	Applied total forces	Resultant
SP2leg	Bilateral standing	[0,0,+400] N on right AC notch and [0,0,+400] N on left AC notch	800 N cranio-caudal
SP1leg	Unilateral standing	[0,0,+800] N on right AC notch	800 N cranio-caudal
LAB1	Internal ring force pair	[+400,0,0] N on ring_contact_left and [-400,0,0] N on ring_contact_right	0 N net; bilateral ML pair
LAB2	Ischial force pair	[+400,0,0] N on left_ischium_tuber and [-400,0,0] N on right_ischium_tuber	0 N net; bilateral ML pair
LAB3	Outlet AP force pair	[0,-400,0] N on pubis_ins and [0,+400,0] N on SCJ	0 N net; AP pair

2.5 Reference configuration and SIJ kinematics

The kinematic analysis compares each unloaded mapped surface $\mathbf{x}_i^0 = \mathbf{X} + \phi_i$ with its loaded surface $\mathbf{x}_{i\ell}^1 = \mathbf{x}_i^0 + \mathbf{u}_{i\ell}$. Archived FE displacement fields were reprocessed without rerunning equilibrium solves; correspondence and pairing checks are documented in Appendix A.1.

Relative ilium-to-sacrum motion was estimated using trimmed Kabsch registration of opposing surface regions within 5 mm, with a minimum of 80 nodes per region. The underlying least-squares rigid fit follows Kabsch (1976); trimming and ROI selection are implementation choices. Before registration, the inverse least-squares affine sacral map was applied to all loaded points. This removes sacral affine strain as well as rigid drift; the resulting quantities are affine-corrected relative motion, rather than unmodified physical joint displacement. The available rigid-only sensitivity check is reported below.

Motion was expressed in a pelvis-aligned ML/AP/CC frame, not a local articular normal/tangent frame. Fixed-axis Cardan xyz angles satisfy $R = R_z R_y R_x$. The bilateral mean of the Euclidean norms of these angle triplets is denoted $|\boldsymbol{\theta}|$; this is a small-angle summary, not the exact invariant rotation angle. Translation $|\mathbf{d}|$ is the bilateral mean of the norms of relative displacements evaluated at the sacral ROI centroid. Component summaries are bilateral

means of absolute values. In particular, $|\theta_x|$ does not distinguish nutation from counternutation, and $|d_{ML}|$ is not a measurement of joint gapping. Translation-magnitude asymmetry is $||\mathbf{d}_L|| - ||\mathbf{d}_R||$.

2.6 Matched variants and variance ratios

The full variant uses subject-specific geometry and bone density. The shape-only variant uses subject geometry and the spatially varying pointwise population-median density field; it is not homogeneous bone. The material-only variant uses subject density on the template geometry. Cartilage and ligament parameters remain nominal in all variants.

For each load and endpoint we calculated $100V_{shape}/V_{full}$ and $100V_{material}/V_{full}$, with 2000 paired subject bootstrap samples (Efron and Tibshirani, 1994). These non-additive ratios can exceed 100% and neither estimate causal variance fractions nor separate geometry–material interactions. Shape/full agreement was also assessed by RMSE, MAE and identity-line $R^2 = 1 - \sum_i (Y_{shape,i} - Y_{full,i})^2 / \sum_i (Y_{full,i} - \bar{Y}_{full})^2$.

2.7 Statistical analysis

The hypotheses organize a retrospective analysis and were not preregistered. The anatomical-adjustment comparison (H3) asked whether including size and pelvic dimensions attenuates age-adjusted female translation differences under localized loading. The matched geometry comparison (H4) asked whether individual geometry retains between-subject variation and individual responses under standardized bone density. Two load-path predictions were that unilateral support increases motion and bilateral asymmetry (H1), and that both transverse force pairs increase the absolute ML component while reducing the CC component relative to bilateral support (H2); the AP pair was assessed separately. Volume–motion associations were exploratory. Distributions are summarized by medians and interquartile ranges. Paired component contrasts use $\Delta_{i\ell,k} = |d_k|_{i\ell} - |d_k|_{i,SP2leg}$, with 3000 bootstrap samples for median confidence intervals and two-sided exact binomial sign tests of the median contrast (zero differences excluded). Wilcoxon signed-rank statistics (Wilcoxon, 1945) are retained in the machine-readable output as a sensitivity diagnostic, but are not used for the primary inference because asymmetric differences can yield a signed-rank result that disagrees with the direction of the median. Benjamini–Hochberg correction was applied to the 12 component comparisons (Benjamini and Hochberg, 1995). H1 additionally uses paired tests for rotation magnitude, translation magnitude and translation

asymmetry (a separate three-test family).

For each LAB case, nested OLS models used HC3 covariance (MacKinnon and White, 1985):

$$\text{M0 : } Y = \beta_0 + \beta_s \text{Female} + \beta_a \text{Age}_z + \epsilon, \quad (3)$$

$$\text{M1 : } Y = \text{M0} + \beta_v \log(V)_z, \quad (4)$$

$$\text{M2 : } Y = \text{M1} + \beta_{AP} \text{AP}_z + \beta_b \text{Biischiadic}_z + \beta_p \text{Subpubic}_z. \quad (5)$$

Here model addition denotes addition to the linear predictor. Female is coded 1 and male 0; continuous predictors are standardized using cohort sample SDs. M2 contains no sex–volume interaction. FDR correction covers its six sex coefficients (two outcomes by three loads). M0/M1 and coefficient attenuation are descriptive. VIFs quantify collinearity without treating a threshold as proof of identifiability. The triad describes inlet depth, interspinous midpelvic width and the anterior pubic arch; it was selected for this retrospective analysis. Pooled and sex-stratified Spearman correlations are exploratory, not evidence of mediation.

Pooled OLS models include load, sex, age, log volume, the morphometric triad, load–volume interactions and a sex–volume interaction, with covariance clustered by the 278 subjects (Cameron and Miller, 2015). A main volume coefficient therefore refers to male SP2leg, not an overall pooled slope. FDR is applied within each outcome’s pooled coefficients. Robust Wald statistics are not converted to partial variance-explained measures.

Exploratory load-specific models regress $\log Y$ on $\log V$, sex, their interaction, age and the triad. Male and female slopes are β_v and $\beta_v + \beta_{v:s}$; HC3 covariance supplies 95% Wald intervals. FDR covers 20 sex-specific slope tests, with a separate family of ten interaction tests. Failure to reject an interaction is not evidence of equal slopes. The alternative $\log(\text{scale})$ model is an algebraic reparameterization because $\log V = 3 \log(\text{scale}) + \log V_{\text{template}}$; it is not an independent robustness test. Sample confidence intervals do not include uncertainty in loads, constitutive choices, segmentation or boundary conditions.

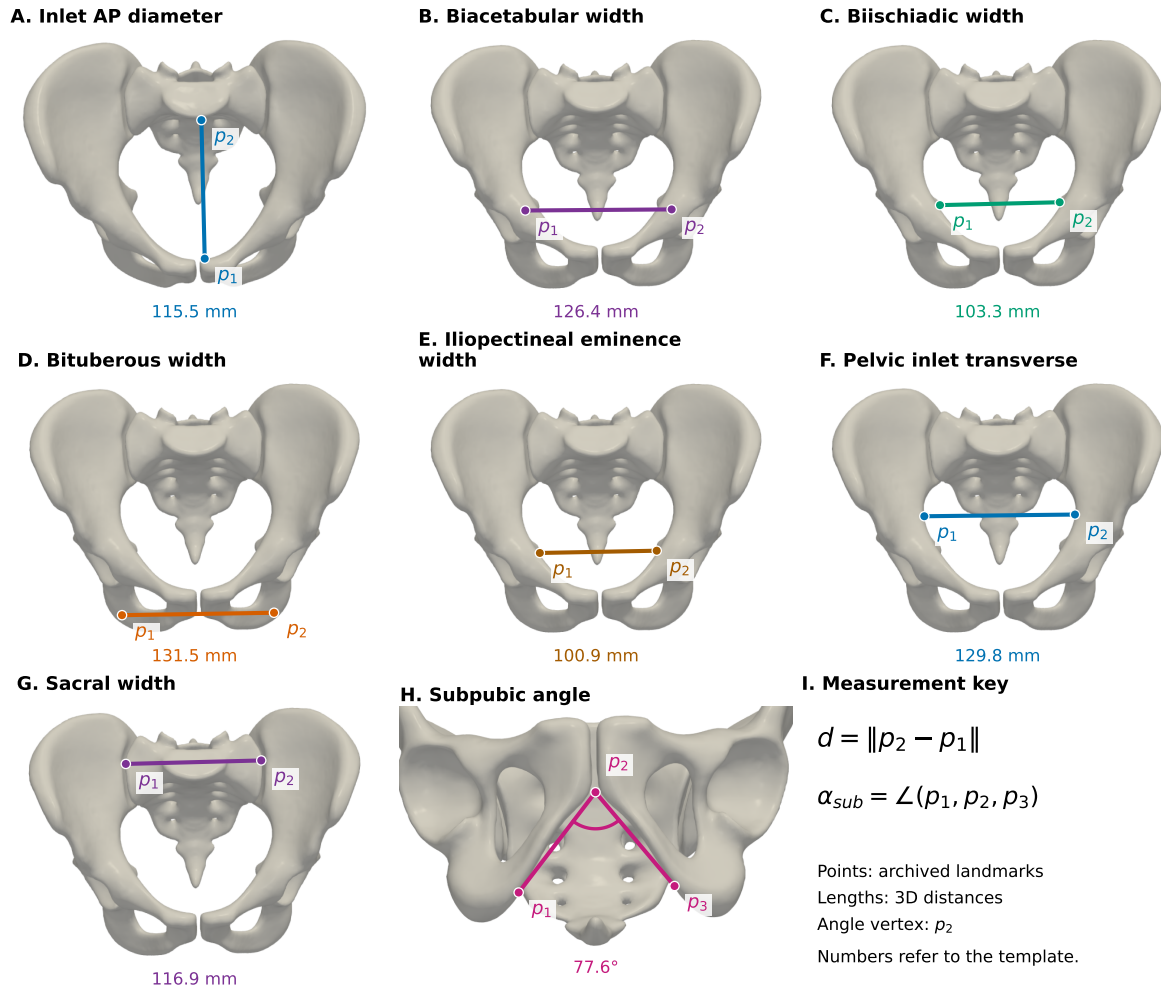


Figure 1: **Anatomical landmarks for all morphometric measures.** (A–G) Archived landmark pairs define the seven linear dimensions in Table 1. (H) The subpubic angle has vertex p_2 and arms toward p_1 and p_3 ; the view is normal to their plane. (I) Measurement definitions and landmark key. Markers and segments are overlaid on the reference pelvic geometry to keep internal landmarks visible. Values describe the template, not cohort averages. Lengths are measured in three dimensions; each camera direction is perpendicular to its measurement segment. Panels use different views and magnifications.

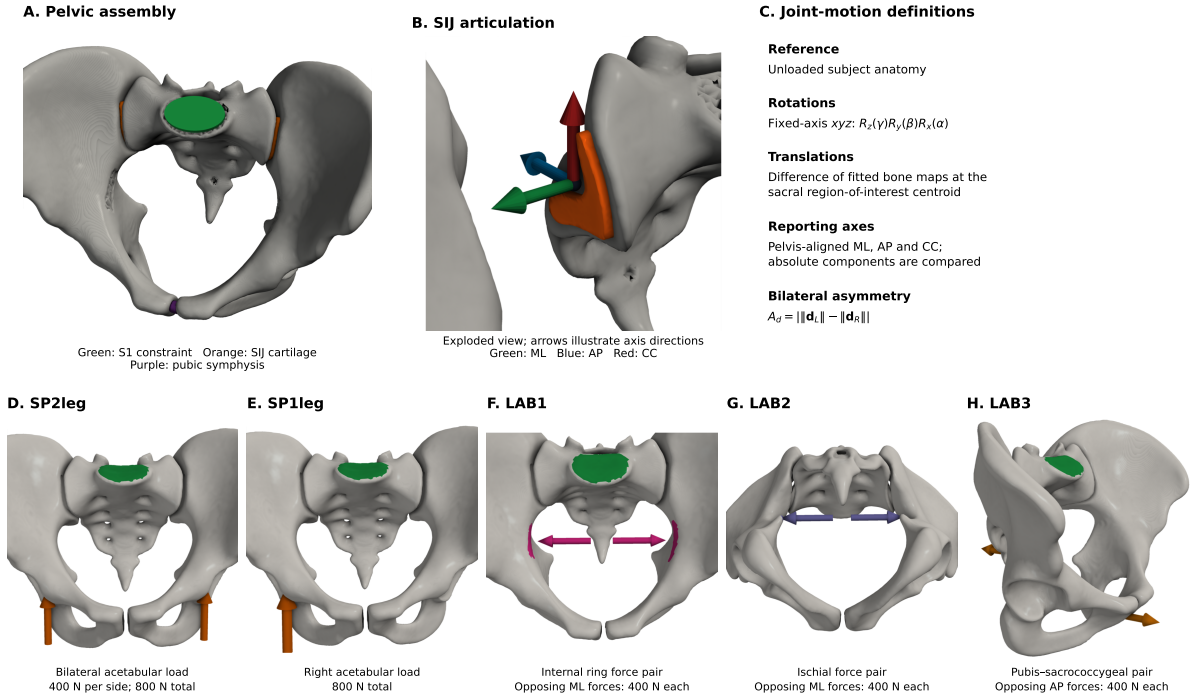


Figure 2: **Anatomy, motion definitions and load application.** (A) Pelvic assembly: green denotes the S1 constraint, orange SIJ cartilage and purple the symphysis. (B) Exploded articulation with illustrative axis directions; these arrows do not define the subject-specific PCA frame or anatomical signs. (C) Rotational and translational reporting conventions. (D,E) Bilateral and unilateral acetabular loads, each totaling 800 N. (F–H) Independent internal-ring, ischial and AP force pairs. Load arrows show applied global directions, not measured joint opening. The S1 penalty approximates a fixed boundary.

Each of the 278 anatomies contributed responses to all five loads (1390 subject–load observations) in each of the three geometry–density variants. This matched design first establishes the loading context for the anatomical comparisons.

3.1 Unilateral support amplifies asymmetry at unchanged total force

Redistributing a fixed total acetabular force changed the SIJ response. Under SP2leg and SP1leg, median rotation norms were 0.956° and 2.040° , and median translations were 1.058 and 1.627 mm, respectively. Median bilateral translation asymmetry was 0.065 mm in SP2leg and 1.602 mm in SP1leg (Fig. 3). The paired SP1leg-minus-SP2leg median difference in rotation norm was $+1.048$ (95% CI $+1.004$ to $+1.102$) degrees, $q < 0.001$. The paired SP1leg-minus-SP2leg median difference in translation magnitude was $+0.582$ (95% CI $+0.559$ to $+0.603$) mm, $q < 0.001$. The paired SP1leg-minus-SP2leg median difference in translation asymmetry was $+1.509$ (95% CI $+1.445$ to $+1.568$) mm, $q < 0.001$. Redistributing the same 800 N total force

220 therefore increased motion and, most prominently, bilateral asymmetry.

Table 3: SIJ motion summaries across 278 subjects. Values are median [interquartile range]; component columns are absolute magnitudes.

Load case	N	Rotation magnitude (deg)	Translation magnitude (mm)	AP translation component (mm)	ML translation component (mm)	CC translation component (mm)
SP2leg	278	0.956 [0.813, 1.109]	1.058 [0.871, 1.207]	0.366 [0.241, 0.482]	0.083 [0.054, 0.123]	0.952 [0.818, 1.112]
SP1leg	278	2.040 [1.765, 2.289]	1.627 [1.430, 1.869]	0.440 [0.357, 0.520]	0.235 [0.164, 0.345]	1.428 [1.213, 1.644]
LAB1	278	0.221 [0.188, 0.268]	0.154 [0.121, 0.190]	0.060 [0.044, 0.079]	0.077 [0.063, 0.092]	0.105 [0.073, 0.144]
LAB2	278	0.724 [0.649, 0.812]	0.255 [0.204, 0.328]	0.176 [0.099, 0.262]	0.134 [0.115, 0.154]	0.075 [0.049, 0.118]
LAB3	278	0.844 [0.672, 1.057]	0.574 [0.448, 0.707]	0.372 [0.260, 0.515]	0.092 [0.069, 0.115]	0.387 [0.326, 0.477]

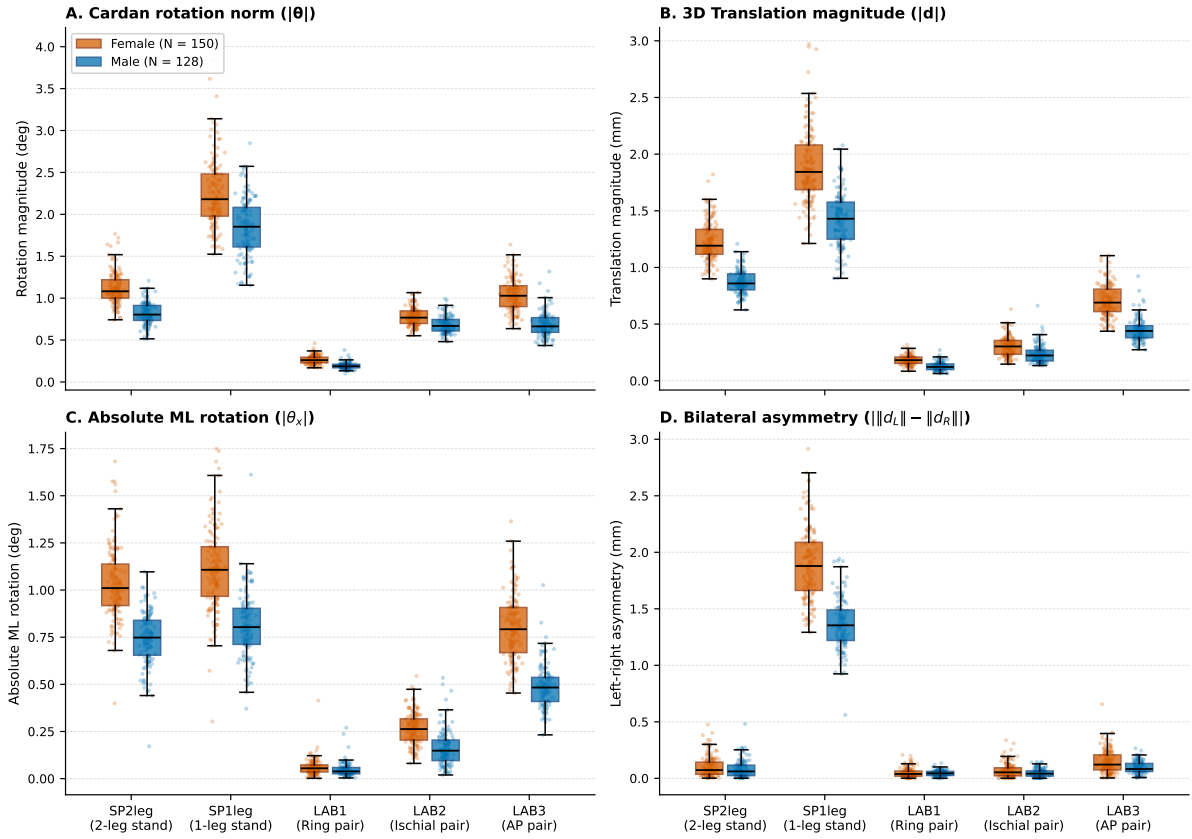


Figure 3: **Load-dependent motion and asymmetry.** Rotation-triplet norm, relative translation norm, absolute ML rotation and bilateral translation-magnitude asymmetry. Points are subjects; boxplots summarize each sex separately using medians and IQRs, with whiskers at 1.5 IQR. The numerical summaries in the text and Table 3 pool both sexes. Female and male groups use vermillion and blue. Jitter is deterministic. No signed nutation direction is inferred.

221 3.2 Load application site selects distinct motion components

222 The localized force pairs separated the response into distinct component patterns (Fig. 4).
 223 Both transverse pairs reduced the absolute CC component relative to bilateral support, but the
 224 ML component increased only under LAB2: its paired median contrast was +0.052 mm (95%
 225 CI +0.047 to +0.061, $q < 0.001$), compared with -0.007 mm for LAB1 (95% CI -0.018 to

+0.002, $q = 0.254$). Both also reduced the AP component. LAB3 increased the ML and AP components while reducing the CC component. The shared transverse-redistribution prediction was therefore only partly supported. Supplementary Table S1 gives all paired contrasts and adjusted probabilities. Component magnitudes quantify motion along the reporting axes; the arrows connect componentwise median contrasts and are not physical trajectories.

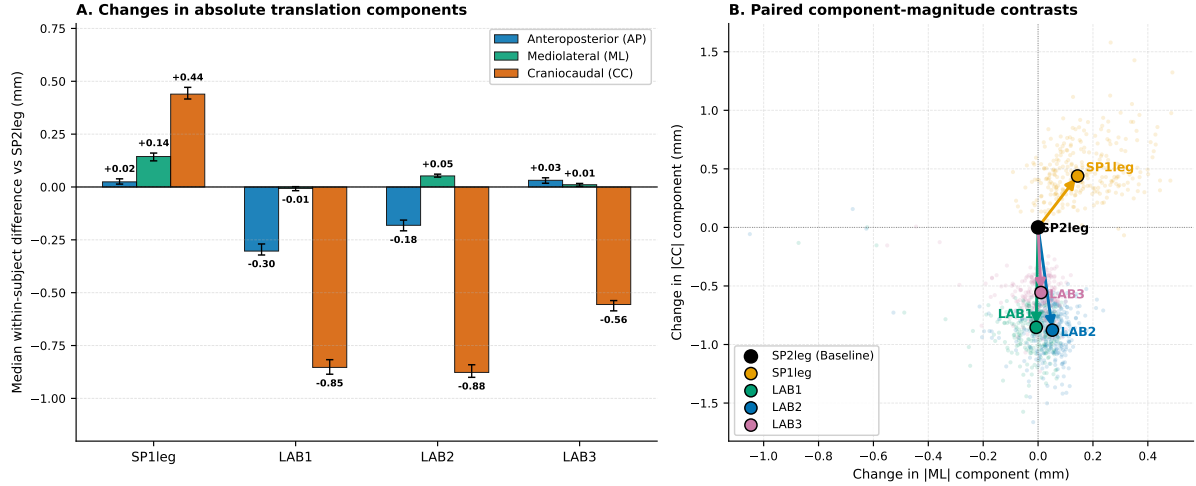


Figure 4: **Paired absolute-component contrasts.** (A) Median within-subject differences from SP2leg, with bootstrap 95% CIs. (B) Individual ML/CC contrast pairs and arrows to their componentwise medians. Positive values mean larger component magnitude, not outward joint motion. Corrected q values are reported in Supplementary Table S1; intervals are pointwise, not simultaneous.

3.3 Pelvic dimensions attenuate sex-associated translation differences

Explicit dimensions substantially attenuated the age-adjusted sex contrast under all three localized loads (Fig. 5). LAB1 female-minus-male translation coefficients were +0.055, +0.040 and +0.013 mm in M0, M1 and M2. The M2 interval was -0.005 to +0.031 mm ($p = 0.166$, $q = 0.566$). LAB2 female-minus-male translation coefficients were +0.071, +0.041 and -0.031 mm in M0, M1 and M2. The M2 interval was -0.078 to +0.016 mm ($p = 0.192$, $q = 0.566$). LAB3 female-minus-male translation coefficients were +0.258, +0.212 and +0.002 mm in M0, M1 and M2. The M2 interval was -0.060 to +0.064 mm ($p = 0.950$, $q = 0.950$). All three fully adjusted translation intervals spanned zero. The maximum non-intercept VIF in M2 was 5.55. Fully adjusted rotation intervals also spanned zero under all three loads (Supplementary Table S3). The exploratory subpubic-angle comparison distinguished the pooled association from those within sex groups. For LAB1, the pooled Spearman correlation between subpubic angle and translation magnitude was $r_s = 0.444$ ($p < 0.001$). The male-only correlation was $r_s = -0.063$ ($p = 0.480$).

The female-only correlation was $r_s = -0.145$ ($p = 0.077$). These exploratory correlations are uncorrected for multiplicity. Figure 5 and Supplementary Table S3 report M2 estimates for both outcomes.

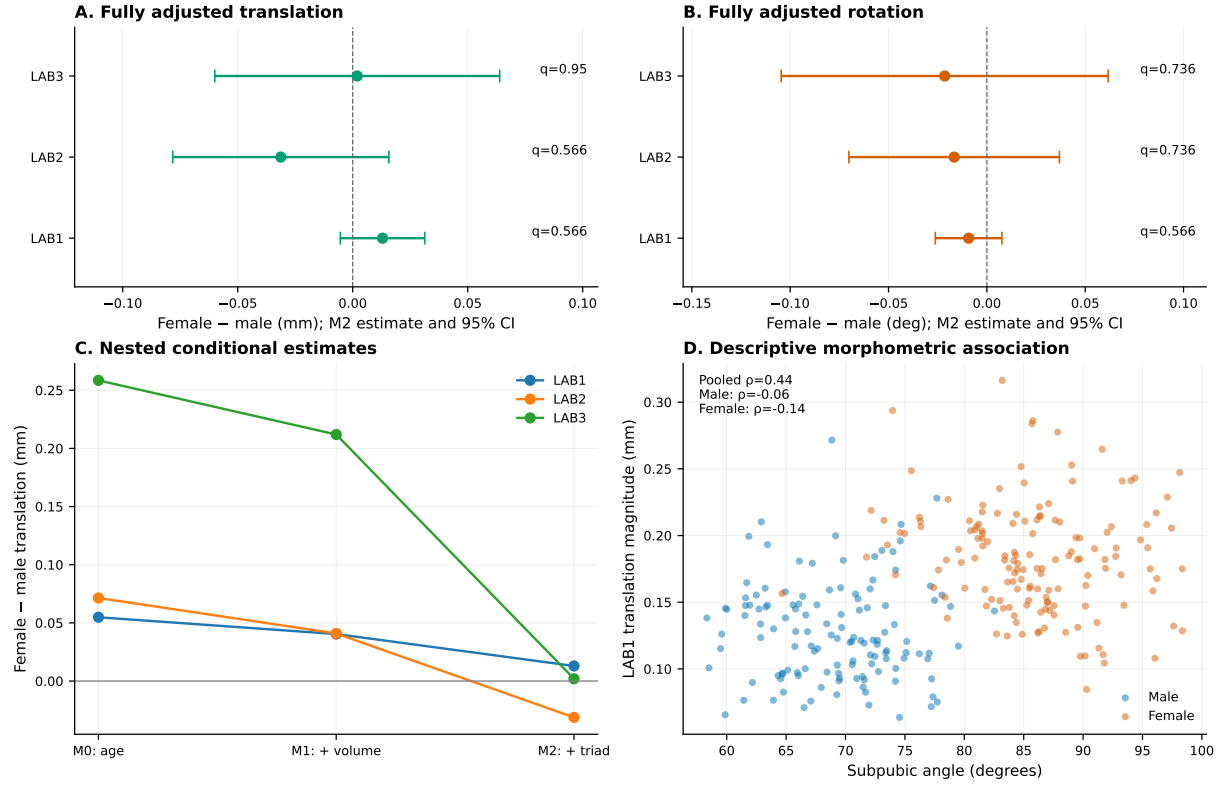


Figure 5: Explicit anatomy attenuates sex-associated translation differences. (A,B) M2 female-minus-male coefficients with HC3 95% CIs, shown on separate translation and rotation axes. Labels give BH-adjusted q values across six tests. (C) Translation coefficients attenuate as volume and dimensions enter the model; connecting lines compare specifications and are not causal pathways. (D) The positive pooled LAB1 subpubic-angle association is not reproduced within either sex. All M2 intervals span zero; this does not establish equivalence. M0 adjusts for age, M1 adds volume, and M2 adds the morphometric triad.

3.4 Individual geometry preserves responses under standardized bone density

Preserving individual geometry while standardizing the bone-density field retained individual motion closely. Across loads, translation had a minimum shape/full identity-line R^2 of 0.9457 and maximum RMSE of 0.0280 mm. For rotation, the corresponding values were 0.8777 and 0.0318 degrees. Across five loads and five endpoints, shape-only/full variance ratios ranged from 96.43% to 120.61% (median 100.68%), compared with 0.014% to 5.789% (median 0.244%) for material-only/full ratios. These are model-variant comparisons, not an additive variance decomposition (Fig. 6; Supplementary Table S2).

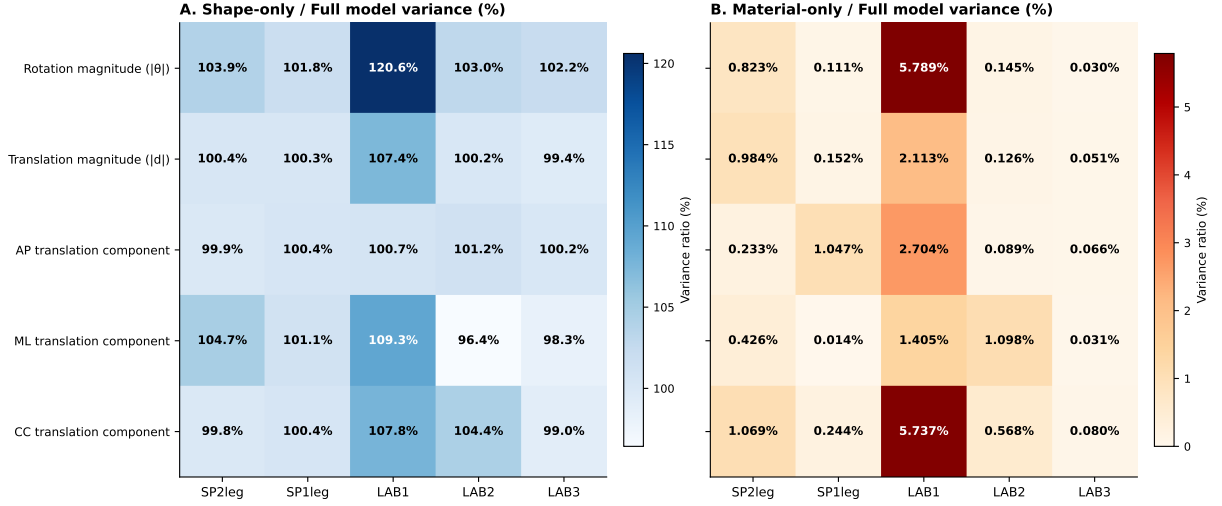


Figure 6: **Retaining individual geometry preserves between-subject variation.** (A) Shape-only/full and (B) material-only/full sample-variance ratios, in percent. Five endpoints are shown. Color scales differ between panels and follow their actual data ranges. Values above 100% are allowed. Subject-level agreement complements these variance summaries (translation RMSE ≤ 0.0280 mm; minimum identity-line $R^2 = 0.9457$). Bootstrap intervals and agreement diagnostics are provided in Supplementary Table S2; neither panel represents variance explained by an independent causal factor.

3.5 Conditional size–motion scaling across loading configurations

Pelvic volume showed a consistent inverse conditional association with motion in the retrospective scaling analysis: all 20 sex-specific slopes were negative and survived FDR correction. Slopes ranged from -1.062 to -0.366; the largest adjusted slope q was 0.002. These estimates describe mechanical response versus pelvic size under identical prescribed forces across subjects within each load configuration, conditional on the other model covariates.

Slope magnitudes varied numerically across load cases (Fig. 7). SP2leg estimates lay numerically near the simple fixed-force dimensional reference values of $-1/3$ for translation and $-2/3$ for rotation; several localized-load estimates were numerically steeper. This comparison is descriptive: the fitted slopes condition on other pelvic dimensions, whereas the references describe uniform geometric scaling. Between-load slope comparisons are descriptive. Table 4 retains the full estimates, intervals and adjusted probabilities.

Sex-by-volume interaction q values ranged from 0.343 to 0.699; nonsignificance does not establish identical conditional slopes. Pooled cluster-robust coefficients are reported in Supplementary Table S5.

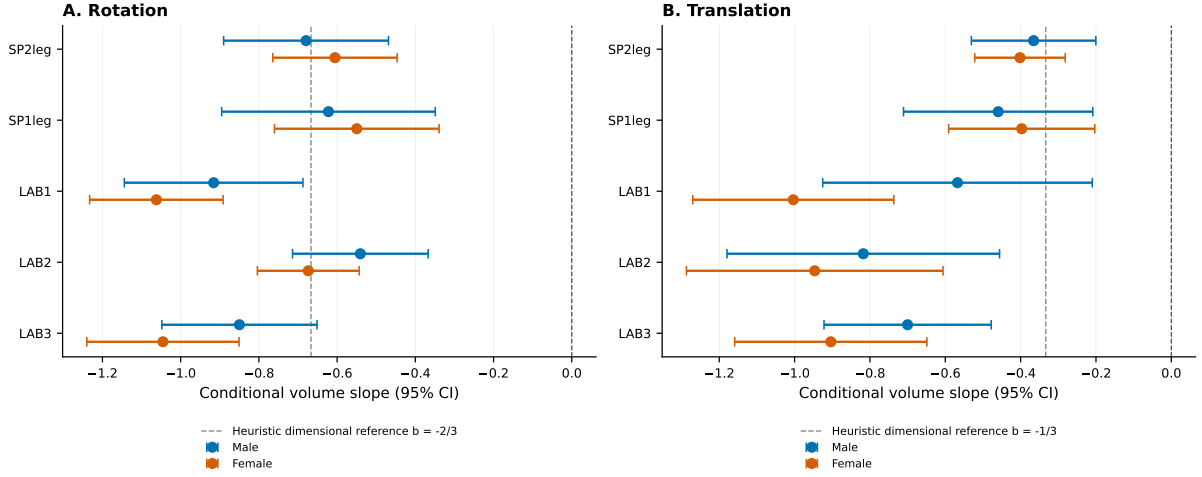


Figure 7: **Conditional log-volume slopes.** Male and female slopes with HC3 95% CIs from the same adjusted log–log models as Table 4. The dashed lines at $b = -2/3$ (rotation) and $b = -1/3$ (translation) are heuristic dimensional references for uniform scaling of geometrically similar linear-elastic structures under fixed force and modulus. They are not fitted values, null hypotheses, expected coefficients of the conditional regression or biological allometric exponents. The regression holds other pelvic dimensions fixed, so numerical proximity to these references is descriptive and does not validate geometric similarity or a scaling law. The zero line indicates no conditional association. Confidence intervals are pointwise; slope q values from the 20-test family are given in Table 4. Raw scatterplots and adjusted conditional geometric-mean curves are supplied as Supplementary Figure S1.

Table 4: Conditional log-volume slopes by load and sex. Slope q values use the 20-test family; interaction q values use a separate ten-test family.

Load	Outcome	Male slope [95% CI]	Male q	Female slope [95% CI]	Female q	Interaction q
SP2leg	Rotation (deg)	-0.679 [-0.890, -0.469]	< 0.001	-0.605 [-0.764, -0.446]	< 0.001	0.699
SP2leg	Translation (mm)	-0.366 [-0.531, -0.200]	< 0.001	-0.402 [-0.522, -0.282]	< 0.001	0.699
SP1leg	Rotation (deg)	-0.622 [-0.895, -0.349]	< 0.001	-0.550 [-0.760, -0.339]	< 0.001	0.699
SP1leg	Translation (mm)	-0.460 [-0.711, -0.208]	< 0.001	-0.397 [-0.591, -0.203]	< 0.001	0.699
LAB1	Rotation (deg)	-0.916 [-1.144, -0.687]	< 0.001	-1.062 [-1.233, -0.891]	< 0.001	0.588
LAB1	Translation (mm)	-0.568 [-0.925, -0.210]	0.002	-1.003 [-1.270, -0.737]	< 0.001	0.343
LAB2	Rotation (deg)	-0.541 [-0.714, -0.367]	< 0.001	-0.674 [-0.804, -0.544]	< 0.001	0.494
LAB2	Translation (mm)	-0.818 [-1.179, -0.456]	< 0.001	-0.947 [-1.287, -0.606]	< 0.001	0.699
LAB3	Rotation (deg)	-0.850 [-1.048, -0.651]	< 0.001	-1.045 [-1.240, -0.851]	< 0.001	0.494
LAB3	Translation (mm)	-0.700 [-0.922, -0.478]	< 0.001	-0.904 [-1.159, -0.649]	< 0.001	0.494

3.6 Sensitivity of motion extraction

In six subjects selected at approximately evenly spaced archive indices (30 subject–load comparisons), replacing affine sacral correction with rigid-only correction changed bilateral rotation norms by at most 0.0229° and translation magnitudes by at most 0.0055 mm. This subset diagnostic does not bound errors in the remaining cohort (Supplementary Table S6).

4 Discussion

Pelvic mechanical response was structured by anatomical scale and individual architecture, and its expression depended strongly on load path. Size showed a consistent inverse conditional relationship with motion across loads. Explicit dimensions substantially attenuated sex-associated translation differences under localized loading, while preserving individual geometry under standardized bone density retained subject-specific responses closely. Redistributing force or changing its application site nevertheless altered the magnitude and component pattern of motion. These observations support an anatomically resolved description of response under specified loading.

Conditional mechanical scaling

The 20 negative sex-specific volume slopes, all retained after multiplicity correction, make size a common mechanical context for the anatomical comparisons. SP2leg slopes lay numerically near the simple dimensional references of $-1/3$ for translation and $-2/3$ for rotation. Several localized-load estimates were numerically steeper, including LAB1 rotation and LAB2 translation; this is a descriptive comparison across loads.

The reference values follow elementary dimensional reasoning: for geometrically similar linear-elastic structures at fixed force F and modulus E , translation scales as $u \sim F/(EL) \propto V^{-1/3}$, and rotation as $\theta \sim u/L \propto V^{-2/3}$. These are heuristic dimensional references. The fitted slopes describe changes in volume at fixed values of the other pelvic dimensions; the dimensional references describe uniform geometric scaling. Why some localized-load estimates are numerically steeper remains unresolved. The pelvis is a jointed structure with heterogeneous tissues and varying proportions. Fixed ligament spring parameters and cartilage properties do not impose geometric similarity across sizes. Geometry-dependent force moments, heterogeneous density fields and conditional adjustment for other dimensions could also contribute to the load-specific slopes. Their contributions remain to be tested.

The slopes describe conditional mechanical scaling under standardized forces. Physiological scaling with body-mass-dependent loading was not evaluated.

Size, shape and sex-associated response

The female and male groups differed in several dimensions: female pelves had lower total bone volume but greater mean inlet depth, interspinous width and subpubic angle (Table 1). Sex groups therefore differ in both size and proportions. Anatomical evidence distinguishes allometric differences in overall proportions from largely non-allometric differences in subpubic angle (Fischer and Mitteroecker, 2017); local auricular shape also differs between sexes (Nishi et al., 2020). Earlier representative-model comparisons of standing SIJ mechanics (Joukar et al., 2018) establish relevant prior evidence, while the present cohort examines how population contrasts change with explicit anatomical adjustment and loading context. Adding those dimensions reduced the female translation coefficients from +0.055 to +0.258 mm across localized loads to -0.031 to +0.013 mm. All three fully adjusted translation intervals under localized loading included zero. Coefficient attenuation quantifies the change in the sex-associated contrast after anatomical adjustment. Sex equivalence and causal mediation were not tested.

The subpubic-angle association depended on whether subjects were pooled or stratified by sex. The positive pooled correlation with LAB1 translation was not reproduced within either sex: both within-sex estimates were weakly negative and their tests did not reject zero correlation. The selected triad represents a subset of pelvic architecture. Out-of-sample predictive performance was not evaluated. External assessment should examine anatomical overlap between sexes, collinearity and performance beyond this cohort.

Geometry retains the subject-specific mechanical response

The matched geometry–density comparison quantified the preservation of individual responses after density standardization. Retaining individual geometry with a spatially varying population-median density field gave translation RMSE no greater than 0.028 mm and a minimum identity-line R^2 of 0.946. The minimum identity-line R^2 for rotation was 0.878. The near-100% shape/full variance ratios and small material-only/full ratios were consistent with these paired comparisons. CT-osteabsorptiometry showed limited differences in mineralization patterns across auricular shape categories, although larger surfaces had higher iliac mineralization (Poilliot et al., 2023). Those observations provide context on long-term subchondral adaptation. Individual geometry may be prioritized when modeling these joint-motion endpoints. The acceptable degree of density simplification depends on the error tolerance.

This hierarchy is conditional on the passive static model and its implemented density–

modulus law, including a low-density modulus floor. Cartilage properties, ligament properties and pretension were fixed. Ligament pretension can substantially affect SIJ predictions (Heyland et al., 2025), and introducing individual soft-tissue variation could change the relative contributions. The results support simplification of the represented bone-density field for the reported joint-motion endpoints under these constitutive assumptions.

Load path determines how architecture is expressed

SIJ mobility depended on force distribution. Redistributing 800 N from bilateral to unilateral acetabular support increased translation and rotation and strongly amplified bilateral asymmetry. The total applied force was unchanged, but its moments and route through the ring changed. This gives a population-level mechanical context to established accounts of SIJ load transfer (Snijders et al., 1993; Vleeming et al., 2012).

Application site was also consequential. Both transverse force pairs reduced the craniocaudal component relative to bilateral support. The mediolateral contrast with bilateral support was positive for the ischial pair, whereas the interval for the internal-ring pair included zero. These comparisons partly supported the shared transverse-redistribution hypothesis and described distinct component patterns; a direct contrast between the two application sites was not reported.

Implications for functional anatomy

Overall size was inversely associated with motion magnitude, preserving individual geometry retained subject-specific responses, and load path altered the expressed motion pattern. Sex captures part of the population variation in anatomical dimensions. Beyond previous work on auricular morphology (Henyš and Hammer, 2025), the present analyses connect whole-pelvis size and dimensions to load-dependent motion and test a geometry-preserving density simplification. The evidence supports describing pelvic function through scale, architecture and loading context together.

Covariation of stature and pelvic shape provides anatomical context for obstetric dimensions (Fischer and Mitteroecker, 2015). The present endpoints quantify relative SIJ motion. Signed joint opening and birth-canal expansion were not measured. Their evaluation under delivery conditions requires direct surface-separation and canal measurements, together with fetal contact and soft-tissue interactions (Chen and Grimm, 2021).

Limitations and experimental tests

The model represents passive, static mechanics with a constrained sacrum, nominal cartilage properties and linearized ligaments, without muscle activation or frictional contact. The density comparison inherits these choices. The cohort spans ages 16–91 and does not specifically represent pregnant or laboring women; pregnancy-related adaptation, parity, pain and delivery outcomes were not analyzed. The retrospective selection of dimensions and comparisons also limits confirmatory interpretation.

Affine sacral correction removes sacral strain as well as rigid drift. The rigid-only sensitivity check showed small endpoint changes in the six subjects examined; its coverage was limited to this subset. Sampling intervals exclude constitutive, anatomical-interpolation and boundary-condition uncertainty. A previous subject-specific pelvic model was validated against cortical strains (Anderson et al., 2005). Physiological SIJ kinematics remain unvalidated in the present cohort. Assessment of mesh and anatomical-interpolation accuracy and independent measurements under matched loads remain necessary, particularly given the difficulty of measuring small SIJ motions (Goode et al., 2008).

The resulting experimental predictions are specific: unilateral loading should increase bilateral asymmetry at matched total acetabular force, and transverse forces at the internal ring and ischial tuberosities should produce different component responses. Tests should measure signed SIJ surface separation alongside the reported point-motion endpoints. Comparing subject-specific and standardized density assignments against those measurements would test the proposed simplification, while introducing measured cartilage, ligament and pretension variation would test its scope. Independent specimens and loading configurations are needed to determine how far the anatomical relationships generalize.

5 Conclusion

Under standardized loading, pelvic size showed a consistent inverse conditional association with SIJ motion, while individual geometry retained much of the subject-specific response under standardized bone density and fixed soft-tissue properties. Load distribution and application site determined how that response was expressed. Explicit pelvic dimensions also substantially attenuated sex-associated translation differences under localized loading. These findings identify anatomical scale, individual architecture and loading context as complementary descriptors of

394 modeled SIJ response.

395 A Mathematical formulation and verification

396 A.1 Archived implementation and correspondence checks

397 Subject landmarks were obtained by thin-plate-spline RBF interpolation of template-to-subject
398 displacements (10 neighbors; smoothing parameter 100). These morphometric interpolation set-
399 tings differ from those used for the FE surface reconstruction. Recomputing the eight measures
400 for all 278 subjects reproduced the archived values within 10^{-8} mm or degrees.

401 Archived displacement fields were reprocessed without new equilibrium solves. Reference-
402 to-reference checks were below 10^{-8} in degrees and mm for every subject. Node ordering was
403 checked against all five archived reference displacement fields. Geometry and displacement were
404 evaluated with the same RBF parameters used by the original workflow. Checkpoints preserve
405 subject row indices consistently across the three variants. Pairing follows the common indexed
406 shape/density inputs and the archived processing order; the auxiliary archives do not contain
407 an independent subject-ID manifest.

408 A.2 Reference mapping and equilibrium

409 Let Ω_0 be the FE template and $\chi_i(\mathbf{X}) = \mathbf{X} + \phi_i(\mathbf{X})$ the unloaded subject geometry. The
410 anatomical registration is bijective; RBF interpolation represents its established displacement
411 map on the FE template. The anatomical mapping and the elastic displacement are different
412 fields. With $\mathbf{F}_i = \mathbf{I} + \nabla_X \phi_i$ and $J_i = \det \mathbf{F}_i$, the physical small-strain tensor is

$$413 \quad \boldsymbol{\varepsilon}_i(\mathbf{u}) = \text{sym}(\nabla_X \mathbf{u} \mathbf{F}_i^{-1}). \quad (6)$$

414 Here $\nabla_X \mathbf{u}$ differentiates the displacement represented on the template; $\text{sym} \mathbf{B} = (\mathbf{B} + \mathbf{B}^T)/2$.
415 Volume integration satisfies $dV_i = J_i dV_0$. Thus, for example, the elastic term on the template
416 is $\int_{\Omega_0} (\mathbf{C}_i \boldsymbol{\varepsilon}_i(\mathbf{u})) : \boldsymbol{\varepsilon}_i(\mathbf{v}) J_i dV_0$. For physical coordinates in mm, find the discrete displacement $\mathbf{u}_h$
417 such that, for every test displacement $\mathbf{v}_h$, the weak problem is

$$418 \quad \int_{\Omega_i} \mathbf{C}_i \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) dV + \int_{\Gamma_{S1,i}} \gamma \mathbf{u} \cdot \mathbf{v} dS + \mathbf{v}_h^T \mathbf{K}_{lig,i} \mathbf{u}_h = \int_{\Gamma_{load,i}} \mathbf{t} \cdot \mathbf{v} dS + \mathbf{v}_h^T \mathbf{f}_{pre,i}. \quad (7)$$

Here $\mathbf{C}_i$ is the isotropic elasticity tensor, $\mathbf{t}$ is the prescribed surface traction, and $\mathbf{u}_h, \mathbf{v}_h$ are nodal coefficient vectors in the spring terms. In each tissue, $\boldsymbol{\sigma} = 2\mu\boldsymbol{\varepsilon} + \lambda \text{tr}(\boldsymbol{\varepsilon})\mathbf{I}$, with $\mu = E/[2(1+\nu)]$ and $\lambda = E\nu/[(1+\nu)(1-2\nu)]$. The surface penalty has units N/mm^3 , because its associated traction is $\gamma\mathbf{u}$ in N/mm^2 . It imposes an approximate fixed boundary, not exact $\mathbf{u} = 0$. The ligament stiffness and pretension terms are required for consistency with the assembled solver. For one spring segment of length L , direction $\mathbf{e}$, assigned stiffness k and pretension T , its directional stiffness is

$$\mathbf{W} = k\mathbf{e}\mathbf{e}^T + (T/L)(\mathbf{I} - \mathbf{e}\mathbf{e}^T). \quad (8)$$

This matrix is assembled with opposite off-diagonal blocks at its two endpoints. It is a linearized spring representation without tension-only switching.

The surface measure obeys $dS_i = J_{surf,i}dS_0$, where

$$J_{surf,i} = J_i \|\mathbf{F}_i^{-T} \mathbf{n}_0\|, \quad \mathbf{t} = \frac{\mathbf{F}^{tot}}{\int_{\Gamma_{load,0}} J_{surf,i} dS_0}. \quad (9)$$

Thus the applied physical force integral equals $\mathbf{F}^{tot}$. The penalty and load integrals use the corresponding mapped surface measures. The force vectors are listed in Table 2; their global axes are not assumed identical to the PCA reporting axes below.

A.3 Unloaded and loaded surfaces

Let $\mathbf{X}_j$ be surface template nodes. The two point clouds used for kinematics are

$$\mathbf{p}_{ij}^0 = \mathbf{X}_j + \phi_i(\mathbf{X}_j), \quad \mathbf{p}_{i\ell j}^1 = \mathbf{p}_{ij}^0 + \mathbf{u}_{i\ell}(\mathbf{X}_j). \quad (10)$$

Both use identical topology and point correspondence. Here “unloaded” denotes the geometric reference with zero stored elastic displacement, not a separately equilibrated pretension-only state. The loaded field includes both the prescribed external load and the ligament pretension right-hand side. Comparing $\mathbf{X}_j$ with $\mathbf{p}_{i\ell j}^1$ would incorrectly include subject morphology in the measured motion. Geometry was reconstructed with the original two-stage interpolation: template-to-FE mapping (10 neighbors, smoothing 50) followed by FE-to-surface mapping (10 neighbors, zero smoothing). Saved elastic fields use the latter interpolation. Cached sparse interpolation weights implement the same linear RBF evaluation; numerical comparisons against direct RBF evaluation check this equivalence.

A.4 Pelvis-aligned reporting frame

The origin $\mathbf{o}$ is the centroid of the unloaded surface. The initial ML axis is the dominant eigenvector of the point-cloud covariance, with deterministic sign. The symmetry-plane routine uses this PCA approximation; it does not optimize reflection symmetry. Project the centered points onto its orthogonal plane and take the leading in-plane singular vector. Orthogonalization gives $\hat{\mathbf{z}}$, followed by

$$\hat{\mathbf{y}} = \frac{\hat{\mathbf{z}} \times \hat{\mathbf{x}}}{\|\hat{\mathbf{z}} \times \hat{\mathbf{x}}\|}, \quad \hat{\mathbf{z}} = \hat{\mathbf{x}} \times \hat{\mathbf{y}}, \quad \mathbf{P} = [\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}]. \quad (11)$$

The ML and AP axes are simultaneously reversed if needed so that ML points from the left to right ilium centroids. This construction is orthonormal and right-handed but does not independently anchor the AP/CC signs to anatomical landmarks. Absolute components are therefore used for comparisons. No clinical nutation sign is inferred from the stored α_x sign.

A.5 Affine correction, regions and rigid fits

Fit the sacral point correspondence by ordinary least squares,

$$(\mathbf{A}_S, \mathbf{b}_S) = \arg \min_{\mathbf{A}, \mathbf{b}} \sum_{j \in S} \|\mathbf{p}_j^1 - \mathbf{A} \mathbf{p}_j^0 - \mathbf{b}\|^2, \quad \mathbf{q}_j = \mathbf{A}_S^+(\mathbf{p}_j^1 - \mathbf{b}_S), \quad (12)$$

where $+$ denotes the Moore–Penrose pseudoinverse. This removes fitted affine sacral deformation, including stretch and shear; it is not only a change of rigid observer frame.

For each side, first restrict both point clouds to that side of the reporting-frame midplane (falling back to the entire bone if this set is empty). Select sacral and iliac points within 5 mm of the opposing restricted point cloud. If fewer than 80 points qualify, use up to the nearest 80 available points. Distances are between discrete points rather than continuous articular surfaces. For each bone ROI, the rigid fit maps $\mathbf{p}^0$ to $\mathbf{q}$. With centered row matrices $\tilde{\mathbf{P}}_B, \tilde{\mathbf{Q}}_B$,

$$\tilde{\mathbf{P}}_B^T \tilde{\mathbf{Q}}_B = \mathbf{U} \Sigma \mathbf{V}^T, \quad \mathbf{R}_B = \mathbf{V} \text{diag}(1, 1, \det(\mathbf{V} \mathbf{U}^T)) \mathbf{U}^T, \quad \mathbf{t}_B = \bar{\mathbf{q}}_B - \mathbf{R}_B \bar{\mathbf{p}}_B^0. \quad (13)$$

The fit is repeated after excluding the largest 10% residuals, using two trimming iterations. Trimming reduces outlier influence but does not validate the ROI as a physical contact patch.

A.6 Relative rotation and translation

For sacral and iliac fits S, I ,

$$\mathbf{R}_{rel} = \mathbf{R}_S^T \mathbf{R}_I, \quad \mathbf{R}_{loc} = \mathbf{P}^T \mathbf{R}_{rel} \mathbf{P} = \mathbf{R}_z(\gamma_z) \mathbf{R}_y(\beta_y) \mathbf{R}_x(\alpha_x). \quad (14)$$

The fixed-axis xyz extraction, for zero-based matrix indices away from gimbal lock, is

$$\alpha_x = \text{atan2}(R_{21}, R_{22}), \quad \beta_y = \arcsin[-\text{clip}(R_{20}, -1, 1)], \quad \gamma_z = \text{atan2}(R_{10}, R_{00}). \quad (15)$$

At $\beta_y = \pm\pi/2$, choose $\gamma_z = 0$ and $\alpha_x = \text{atan2}[-\text{sign}(R_{20})R_{01}, R_{11}]$. In implementation, the equivalent stable expression $\beta_y = \text{atan2}(-R_{20}, \sqrt{R_{00}^2 + R_{10}^2})$ is used, with the singular branch restricted to vanishing cosine. The singular branch is included for completeness; physiological small-angle outputs should remain far from it. Matrix reconstruction tests cover both branches.

For the unloaded sacral ROI centroid $\mathbf{p}_c$, the reported displacement is

$$\mathbf{d} = \mathbf{P}^T [(\mathbf{R}_I - \mathbf{R}_S) \mathbf{p}_c + \mathbf{t}_I - \mathbf{t}_S]. \quad (16)$$

It evaluates the difference of two fitted maps at a common reference point, expressed in the unloaded pelvic axes after affine correction. It is not the translational parameter of $T_S^{-1}T_I$. In fact, with $\mathbf{t}_{rel} = \mathbf{R}_S^T(\mathbf{t}_I - \mathbf{t}_S)$,

$$\mathbf{d} = \mathbf{P}^T \mathbf{R}_S [(\mathbf{R}_{rel} - \mathbf{I}) \mathbf{p}_c + \mathbf{t}_{rel}]. \quad (17)$$

This identity makes explicit the distinction between the relative rigid transform and the reported point-displacement difference. It is neither the distance between opposing articular surfaces nor a displacement along a surface normal. The bilateral summaries are

$$|\boldsymbol{\theta}| = \frac{1}{2} \sum_{s=L,R} (\alpha_{x,s}^2 + \beta_{y,s}^2 + \gamma_{z,s}^2)^{1/2}, \quad |\mathbf{d}| = \frac{1}{2} \sum_{s=L,R} \|\mathbf{d}_s\|, \quad (18)$$

$$|d_k| = \frac{1}{2} (|d_{k,L}| + |d_{k,R}|), \quad A_d = \left| \|\mathbf{d}_L\| - \|\mathbf{d}_R\| \right|. \quad (19)$$

Angles are converted to degrees before reporting. The exact principal rotation angle would instead be $\arccos[(\text{tr } \mathbf{R}_{rel} - 1)/2]$ with numerical clipping. The Cardan norm approximates it at small angles and depends on the chosen axes and sequence.

A.7 Conditional size scaling

The archived size-measurement workflow sums volumes of the selected left ilium, right ilium and sacrum surface bodies, using mapped surface geometry and converting mm^3 to cm^3 . These are whole-bone geometric volumes, not birth-canal volume or a direct integral of cortical tissue. The size factor is $s_i = (V_i/V_{\text{template}})^{1/3}$. Therefore a log-volume slope b becomes $3b$ under log scale, with identical fitted responses and tests.

For the sex-interaction log model, $b_M = \beta_v$ and $b_F = \beta_v + \beta_{v:s}$. The female slope variance is

$$\text{Var}(b_F) = \text{Var}(\beta_v) + \text{Var}(\beta_{v:s}) + 2 \text{Cov}(\beta_v, \beta_{v:s}). \quad (20)$$

Reported intervals use this full covariance. Exponentiating a fitted log response gives a conditional geometric mean, not an unbiased arithmetic mean. Figure 7 reports slopes from this model. Supplementary Figure S1 shows its conditional geometric-mean curves with age and the morphometric covariates set to their cohort means.

A.8 Statistical estimands and uncertainty

All logarithms are natural logarithms of numerical values in the stated units: $\log V$ abbreviates $\log[V/(1 \text{ cm}^3)]$, and $\log Y$ uses 1 mm or 1 degree as the reference unit. Changing these reference units changes intercepts, but not the sex-specific volume slopes. Standardized predictors are $z_i = (x_i - \bar{x})/s_x$, where s_x is the cohort sample standard deviation; for log volume, the logarithm is taken before standardization.

For design matrix $\mathbf{Z}$, OLS residuals e_i and leverage h_{ii} , the HC3 covariance estimator is

$$\widehat{\text{Cov}}_{\text{HC3}}(\hat{\beta}) = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \text{diag} \left[\frac{e_i^2}{(1 - h_{ii})^2} \right] \mathbf{Z} (\mathbf{Z}^T \mathbf{Z})^{-1}. \quad (21)$$

For any coefficient contrast $\mathbf{c}$, its estimate is $\mathbf{c}^T \hat{\beta}$ and standard error is $[\mathbf{c}^T \widehat{\text{Cov}}_{\text{HC3}} \mathbf{c}]^{1/2}$. Reported Wald intervals use the normal critical value (approximately 1.96). The volume slopes are conditional elasticities: holding the other predictors fixed, multiplying volume by $a > 0$ multiplies the fitted geometric-mean response by a^b .

For a paired contrast Δ_i , the reported effect is $\text{median}_i \Delta_i$, not the difference between two marginal medians. With n_* nonzero differences and $B = \sum_i \mathbb{I}(\Delta_i > 0)$, the two-sided exact sign test uses $B \sim \text{Binomial}(n_*, 1/2)$ under the null of balanced signs. For a continuous dif-

ference distribution this is a test of zero median; excluding exact zeros conditions on nonzero differences. Bootstrap intervals are percentile intervals obtained by resampling subjects, preserving the paired observations. They are pointwise intervals, whereas the reported q values apply Benjamini–Hochberg adjustment within the specified test families; these two summaries need not cross their respective thresholds together.

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Author contributions

CRedit authorship contribution statement. Petr Henyš: Conceptualization; Methodology; Software; Formal analysis; Investigation; Data curation; Visualization; Funding acquisition; Writing – original draft; Writing – review & editing. Niels Hammer: Supervision; Writing – review & editing.

Conflict of interest statement

The authors declare no conflicts of interest.

Supporting information

Supporting Information provides statistical definitions, component contrasts, variance ratios, sex models, the scale reparameterization, and extraction sensitivity diagnostics.

Data Availability Statement

The manuscript working package contains corrected subject-level endpoints, statistical tables and figures. The original FE displacement archives are retained separately.

Code Availability Statement

The working package includes scripts for corrected endpoint extraction, statistical analysis and figure generation. The project development repository is [available on GitHub](#).

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